

# Disproof of conjecture 00000001752

earthking11

2026-09-14

## Verdict: FALSE

We disprove conjecture 00000001752 as stated, in its own Burnside-average reading. The refutation is unconditional, elementary, and finite: for  $S_3$  acting (sharply) 2-transitively on the 1-subsets of  $\{1, 2, 3\}$  the Burnside average of fixed-point counts is 1, whereas the conjectured closed form gives  $1!S(3, 1)/6 = 1/6$ , and  $1 \neq 1/6$ .

**Scope statement (read this first).** We refute the definite equation that the conjecture file itself writes in parentheses, namely that the vector of Mark values *equals the Burnside average of subset counts*,  $\frac{1}{|G|} \sum_{g \in G} \text{Fix}_k(g) = k! S(n, k)/|G|$ . Two readings must be kept apart:

- **In scope.** The Burnside-average equation, with the right-hand side exactly as the file states it,  $k! S(n, k)/|G|$  and  $|G| = n!$ . Our decisive witness  $S_3$  on 1-subsets lies inside the strict 2-transitive case. This is what we disprove.
- **Out of scope.** Any action that is *not* 2-transitive. We use  $S_4$  on 2-subsets only as an additional illustration of the identity being false, and we explicitly flag it as outside the 2-transitive case: its ordered pairs of distinct 2-subsets split into intersecting and disjoint orbits, so it is not 2-transitive.

We also state plainly the honest caveat that the conjecture is auto-generated and misdefines “Mark”; see Section 5.

## 1 The conjecture, quoted verbatim

From `conjectures/00000001752.md`:

**English.** Definition: The Mark values of primitive permutation characters of  $S_n$  are the averages of fixed points (by subset size). Conjecture: The Mark values in the 2-transitive case have a complete closed form:  $M_k = (k! \cdot S(n, k))/|G|$  (given by the Burnside average of subset counts); agreeing case-by-case with the ATLAS.

The conjecture is filed in both English and Chinese; the two versions are identical in content. The PDF’s Latin Modern fonts do not embed CJK glyphs, so the Chinese text is kept verbatim (with embedded Unicode) in `README.md` rather than duplicated here, following the same convention as the reference submission for 00000000463. In both languages the parenthetical identifies the right-hand side as *the Burnside average of subset counts*, which is the equation we take as the formal content and disprove.

## 2 The decisive in-scope witness: $S_3$ on 1-subsets

Take  $n = 3$ ,  $k = 1$ , and  $G = S_3$  acting on the three 1-subsets  $\{1\}, \{2\}, \{3\}$  of  $\{1, 2, 3\}$ . This is the natural action on three points, which is sharply 2-transitive:  $|G| = 3 \cdot 2 = 6$ . It is therefore squarely inside the strict “2-transitive case” of the conjecture.

Write  $\text{Fix}_1(g)$  for the number of 1-subsets fixed by  $g$ ; a 1-subset  $\{i\}$  is fixed by  $g$  iff  $g(i) = i$ . The six elements of  $S_3$  and their fixed-point counts are:

| $g$         | cycle type | $\text{Fix}_1(g)$ |
|-------------|------------|-------------------|
| $e$         | $1^3$      | 3                 |
| $(1\ 2)$    | $2\ 1$     | 1                 |
| $(1\ 3)$    | $2\ 1$     | 1                 |
| $(2\ 3)$    | $2\ 1$     | 1                 |
| $(1\ 2\ 3)$ | $3$        | 0                 |
| $(1\ 3\ 2)$ | $3$        | 0                 |

Thus the multiset of fixed-point counts is

$$[\text{Fix}_1(g) : g \in S_3] = [3, 1, 1, 1, 0, 0], \quad \sum_{g \in S_3} \text{Fix}_1(g) = 3 + 1 + 1 + 1 + 0 + 0 = 6.$$

**Left-hand side (Burnside average).**

$$\frac{1}{|G|} \sum_{g \in G} \text{Fix}_1(g) = \frac{6}{6} = 1.$$

This is not an accident of the computation: the action of a group on a set is transitive iff it has a single orbit, and Burnside’s lemma says the average number of fixed points equals the number of orbits.  $S_3$  is transitive on 1-subsets, so the average is exactly 1.

**Right-hand side (the conjectured closed form).** With  $n = 3$ ,  $k = 1$ ,  $S(3, 1) = 1$  (the only partition of a 3-set into one block),  $1! = 1$ , and  $|G| = 3! = 6$ :

$$\frac{k! S(n, k)}{|G|} = \frac{1! \cdot 1}{6} = \frac{1}{6}.$$

**Conclusion.** The two sides are 1 and  $1/6$ :

$$\frac{6}{6} = 1 \neq \frac{1}{6}.$$

Since the two fractions have the common denominator 6, this is the integer inequality  $6 \neq 1$ ; equivalently, cross-multiplying  $1 = 1/1$  against  $1/6$ , we compare  $1 \cdot 6 = 6$  with  $1 \cdot 1 = 1$  and find  $6 \neq 1$ . Hence the conjectured equation fails for  $S_3$  on 1-subsets, an action that *is* 2-transitive. The refutation is in scope.

*Remark 1* (the  $k = 2$  case of  $S_3$  is the one coincidence). For  $n = 3$ ,  $k = 2$ , the 2-subsets are the complements of the singletons, so  $S_3$  acts on them exactly as it does on the singletons:  $\text{Fix}_2$  is again  $[3, 1, 1, 1, 0, 0]$ , the Burnside average is again 1, and  $2! S(3, 2)/6 = 2 \cdot 3/6 = 1$ . Here the two sides agree. This is the isolated coincidence  $(n, k) = (3, 2)$ ; it does not rescue the conjecture, which fails at  $(3, 1)$  and, as we now see, almost everywhere else.

### 3 The general 2-transitive range

**Theorem 2.** *Let  $n \geq 2$  and  $1 \leq k \leq n - 1$ . The action of  $S_n$  on the  $k$ -subsets of  $\{1, \dots, n\}$  is 2-transitive if and only if  $k = 1$  or  $k = n - 1$ . (For  $n = 3$  both conditions hold at  $k = 2 = n - 1$ , so the case  $(S_3, 2)$  is already covered; there is no further exception.)*

*Sketch.*  $S_n$  is transitive on  $k$ -subsets for every  $k$ , and it is 2-transitive on 1-subsets (the natural action) and on  $(n - 1)$ -subsets (complements). For  $2 \leq k \leq n - 2$  the intersection size of two distinct  $k$ -subsets is preserved by every permutation, but not all values  $0, 1, \dots, k - 1$  occur independently: for  $n \geq 4$  there exist disjoint and intersecting pairs, which lie in different orbits; hence the action is not 2-transitive. For  $n = 3$  the only  $k$  with  $2 \leq k \leq n - 2 = 1$  is empty, consistent with the statement.  $\square$

On this range the two sides have the following closed forms.

**Proposition 3.** *For  $n \geq 2$ :*

$$\frac{1}{|S_n|} \sum_{g \in S_n} \text{Fix}_k(g) = 1 \quad \text{for all } 1 \leq k \leq n - 1,$$

*while the conjecture's right-hand side is*

$$\frac{k! S(n, k)}{n!} = \begin{cases} \frac{1}{n!}, & k = 1, \\ \frac{n - 1}{2}, & k = n - 1 \end{cases} \quad (1 \leq k \leq n - 1).$$

*Proof.* The left-hand side is the number of orbits of  $S_n$  on  $k$ -subsets, which is 1. For  $k = 1$  we have  $S(n, 1) = 1$ , so  $1! S(n, 1)/n! = 1/n!$ . For  $k = n - 1$ ,  $S(n, n - 1) = \binom{n}{2}$ , so

$$\frac{(n - 1)! S(n, n - 1)}{n!} = \frac{(n - 1)! \binom{n}{2}}{n!} = \frac{\binom{n}{2}}{n} = \frac{n - 1}{2}. \quad \square$$

**Corollary 4** (failure throughout the 2-transitive range). *For  $n \geq 2$  the conjecture's equation holds in the 2-transitive range only at the isolated coincidence  $(n, k) = (3, 2)$ . In particular:*

- at  $k = 1$  the right-hand side is  $1/n! \neq 1$  for every  $n \geq 2$ ;
- at  $k = n - 1$  the right-hand side is  $(n - 1)/2 \neq 1$  for every  $n \geq 4$ ;
- the only other value,  $k = n$  (a single  $k$ -subset), is not 2-transitive and both sides equal 1 there.

The first bullet is already decisive at  $(n, k) = (3, 1)$ , and it fails for *every*  $n \geq 2$ , not merely generically.

### 4 The formula is not integral, so it cannot be a Mark value

A Mark in the ATLAS sense is a permutation-character value, i.e. a number of fixed points of a group element, hence a non-negative *integer*. The conjectured right-hand side is frequently *not* an integer. On the in-scope range:

- at  $k = 1$  (in scope for every  $n \geq 2$ ) the value is  $1/n!$ , which is a non-integer for every  $n \geq 2$ :  $1/2, 1/6, 1/24, 1/120, \dots$ ;

- at  $k = n - 1$  the value  $(n - 1)/2$  is a non-integer for every even  $n$  (e.g.  $n = 4$  gives  $3/2$ ,  $n = 6$  gives  $5/2$ );
- outside the 2-transitive range, e.g.  $S_4$  on 2-subsets, the value is  $2! S(4, 2)/4! = 2 \cdot 7/24 = 7/12$  (this case is out of scope, and is listed only as an illustration).

Meanwhile the quantity it is equated to, the Burnside average, is 1 — an integer. So even before numerical comparison, a fractional right-hand side cannot be “the Mark values”, which are integers. We present this as a secondary, independent obstruction; the primary refutation is the exact numerical mismatch of Section 2.

## 5 Honest caveat: the definition of “Mark” is ill-posed

We state this prominently and do not hide it. The conjecture is auto-generated and its first sentence conflates two different objects:

1. the *permutation character value*  $\pi_k(g) = \text{Fix}_k(g)$ , i.e. the number of  $k$ -subsets fixed by  $g$  — this is the classical Mark value, and it is an integer; and
2. the *average*  $\frac{1}{|G|} \sum_{g \in G} \text{Fix}_k(g)$ , which by Burnside equals the number of orbits (here 1), and is a rational number whose value happens to be integral here.

The first sentence says the Mark values “are the averages of fixed points”, identifying (1) with (2); the parenthetical then says the right-hand side is “given by the Burnside average of subset counts”, which is (2). So:

- **The equation we refute is definite and is the file’s own.** Under the parenthetical reading, the asserted identity is  $\frac{1}{|G|} \sum_g \text{Fix}_k(g) = k! S(n, k)/|G|$ , and it is false (Section 2). This is the reading in which “ $M_k$ ” is the Burnside average, and here our verdict is unconditional.
- **Under the strictest reading, the statement is not well-posed.** If “ $M_k$  is an ATLAS Mark” is taken literally, then  $M_k$  is an integer character value, and equating it to a Burnside average (or to a fraction such as  $1/6$ ,  $7/12$ ,  $1/24$ ) is a type error: an average is not a Mark. In that reading the conjecture is ill-typed rather than false, and we say so plainly rather than pretend otherwise.

Either way the conjecture as filed cannot stand: the definite Burnside-average equation it writes is false, and the strict Mark reading is not a well-posed statement. We do not claim more than this.

## 6 Reproducibility

**Python (standard library only).**

```
$ python3 reproduce.py ⇒ PASS, exit 0
```

The script enumerates all  $3^3 = 27$  functions  $\{0, 1, 2\} \rightarrow \{0, 1, 2\}$  and keeps the 6 bijections; computes  $\text{Fix}_k$  for  $k = 1, 2$  setwise; forms the Burnside sums and averages as exact `fractions.Fraction` values; cross-checks  $k! S(n, k)/n!$  against a self-written Stirling recurrence ( $S(, )$  table and  $S(n, k) = k S(n - 1, k) + S(n - 1, k - 1)$ ); asserts  $1 \neq 1/6$  by cross-multiplication ( $6 \neq 1$ ); and verifies which  $(n, k)$  actions with  $2 \leq n \leq 7$  are 2-transitive by counting orbits on ordered pairs of distinct  $k$ -subsets. It prints `PASS` and exits 0 exactly when every check holds. Observed output includes

$$\text{Fix}_1 = [3, 1, 1, 1, 0, 0], \quad \sum_g \text{Fix}_1(g) = 6, \quad \text{average} = 1, \quad \text{formula} = \frac{1}{6}.$$

## LaTeX.

```
$ tectonic --outdir build main.tex
```

produces build/main.pdf.

## Lean 4.

```
cd lean4
lake build ⇒ exit 0
lake env lean Check.lean
```

Core Lean only (import Std, no Mathlib, no sorry, no native\_decide); Check.lean prints #print axioms and reports no sorryAx. Observed audit:

```
'Tlmc1752.perms_length' depends on axioms: [propext]
'Tlmc1752.burnsideSum_eq' depends on axioms: [propext]
'Tlmc1752.formula_val' does not depend on any axioms
'Tlmc1752.burnside_average_is_one' depends on axioms: [propext]
'Tlmc1752.mismatch' depends on axioms: [propext]
'Tlmc1752.one_ne_one_sixth_cross' does not depend on any axioms
```

## 7 Formalisation

In lean4/Main.lean we formalise the core of Section 2 in core Lean. Since Nat.factorial and Finset are absent from import Std, we hand-roll fact and stirling2, enumerate  $S_3$  as the eraseDups.length == 3 functions among all 27 on Fin 3, compute Fix<sub>1</sub> as List.countP (fun s => f s == s), and prove:

```
theorem perms_length : perms.length = 6
theorem burnsideSum_eq : burnsideSum = 6
theorem formula_val : fact 1 * stirling2 3 1 = 1
theorem burnside_average_is_one : burnsideSum = perms.length
theorem mismatch : burnsideSum ≠ fact 1 * stirling2 3 1
theorem one_ne_one_sixth_cross : (1 : Nat) * 6 ≠ (1 : Nat) * 1
```

Both sides share the denominator  $|G| = 6$ , so the mismatch is the Nat inequality  $6 \neq 1$ ; no Rat and no decide on rationals is needed (which is fortunate, since decide cannot reduce Rat). The concrete evaluations are kernel computations (rfl), and the audit shows only propext for the four theorems that need it and no axioms at all for the other two. See lean4/README.md.

## Status against the submission rules

Rule 3 requires a LaTeX source, a PDF document, and a Lean 4 project.

- **LaTeX source** — main.tex, a standalone article using geometry, amsmath, amssymb, amsthm, array, parskip, booktabs; compiles with tectonic --outdir build main.tex.
- **PDF** — at build/main.pdf, produced by that command.

- **Lean 4 project** — `lean4/`, with the toolchain `leanprover/lean4:v4.33.1` in `lean-toolchain`, a `lakefile.toml` (library `tlmc1752`, no dependencies), `Main.lean`, `Check.lean` (`#print axioms`), and `lean4/README.md`. `lake build` exits 0 and the audit reports no `sorryAx`.